

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

☎ +49 176 62145797 • ✉ a01750283@tec.mx • [LinkedIn](#)

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience collecting, validating, and reconciling regulated data at Deutsche Bundesbank, then reporting findings clearly to both technical and non-technical stakeholders. Comfortable documenting and improving data workflows end-to-end, from automated checks through to structured, auditable reporting. Builds on a visualization background (Tableau, Matplotlib, Seaborn) and is quick to pick up new BI tooling such as Power BI.

Core Competencies

- **Data Validation & Reconciliation:** Automated migration and error-detection testing on regulatory data at Deutsche Bundesbank (row counts, referential integrity, aggregate consistency)
- **Governance & Reporting:** Automated ticketing and stakeholder notification workflows; structured, auditable documentation of data-quality findings
- **Process Improvement:** Converted a non-scalable, case-by-case validation script into a parameterized, reusable, configuration-driven solution
- **Data Visualization:** Tableau, Matplotlib, Seaborn; comfortable learning new BI tools (Power BI) quickly given existing visualization foundation
- **Databases & Tools:** SQL, SQLite, MySQL, Excel, Git, Jira
- **Languages:** English (C1, TOEFL iBT 105/120), German (C1), Spanish (native)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Significantly reduced response time to data quality issues; worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)
- **Automated Validation Framework for Regulatory Reporting Data** – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution
 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Projects

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log documenting every change (table, column, old/new value, timestamp); 5 quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks; built predictive models (Random Forest, Neural Networks) and K-Means clustering for high-risk segmentation
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.